# Language Development

*PSYC 3369 | Fall 2026*

## 1. General Information

### Course Info

**Meeting Time:** Tuesdays, 4:10–6:00 PM

**Location:** 237 Milbank Hall

**Course Website:** CourseWorks link

**Instructor:** Prof. Tyler Knowlton

**Email:** tknowlto@barnard.edu

**Office:** 415 I Milbank Hall

**Office Hours:** Thursdays 2:30–4:15 (or by appointment)

### Course Description

Newborn humans lack the ability to control their limbs and support their head, and can hardly see what's in front of them. They certainly can't talk or understand what's being said to them. But in the span of 3-4 years, they'll transform into linguistically sophisticated creatures, capable of telling stories and conversing with adults. This course explores how that happens. To do so, we'll bring together what's known about linguistic theory, cognitive development, and learning in biological organisms. Along the way we'll discuss methods used to study language acquisition and development (e.g., habituation-switch; preferential looking; truth-value judgement tasks; corpus analysis; human simulation paradigm) and foundational concepts in the field (e.g., data- vs. knowledge-driven learning; nativism and empiricism; poverty of the stimulus arguments). Specific topics will include: word learning, phonological and syntactic categories, syntactic, semantic, & prosodic bootstrapping, quantifier scope, and question formation.

Intro Psych, Intro Ling, or Intro CogSci are officially listed as prerequisites. In practice, rudimentary knowledge of syntax and/or phonology as well as an understanding of basic experimental design would go a long way, but none are strictly required.

### Learning Objectives

- Read, understand, and critically evaluate experimental research on infant and child language.
- Demonstrate knowledge of some of the key debates that have shaped the field of child language acquisition and development.
- Communicate research findings effectively in both written form and in oral presentations (i.e., be able to summarize a paper the way developmental scientists would in a lab meeting).
- Design a novel experiment addressing an open question in some area of language development.

## 2. Materials and Format

### Course Materials

This course will use a mix of primary readings (journal articles) and chapters from Lidz & Knowlton (2026) *A course in first language acquisition*. You can obtain a virtual copy of the book from Columbia University Libraries by searching CLIO or clicking here (make sure you're logged in with your Barnard/Columbia account then click the "Oxford academic" link on the right and you'll be able to download the PDF of any chapter). All journal articles will be posted as PDFs on the class CourseWorks page (under readings).

CourseWorks is also where all assignments should be submitted and where other communications (e.g., announcements, syllabus changes, grades) will be housed.

### Class Format

This is an in-person seminar meeting once each week. The course is discussion-driven, so come having done the reading and ready to talk about it! Starting after the first month, each class also includes student presentations of an assigned research article (see Requirements & Evaluation below for details).

We will also have one meeting outside of class to observe children playing at the Barnard Center for Toddler Development.

## 3. Requirements & Evaluation

Here's a breakdown of what will go into your final grade:

| Component | Weight |
|---|---|
| Article presentation | 15% |
| Toddler Development Center reflection | 10% |
| Final project – Topic pitch | 5% |
| Final project – Project proposal | 10% |
| Final project – Annotated bibliography | 15% |
| Final project – Final paper | 30% |
| Participation | 15% |

### Article presentation (15%)

You will be asked to lead the class discussion of a paper one time during the semester (not for the entire duration of the class, just for as long as it takes to discuss that paper). Available papers/dates are flagged in the course schedule below. You can sign up for a particular paper on CourseWorks (under assignments/article presentation). There should be more papers than there are students enrolled, so hopefully everyone can pick one that best matches their interests.

You should prepare slides that summarize the main point of the paper and the theoretical issue(s) at stake, the particular hypothesis or hypotheses under consideration, and the evidence/argument presented. Your slides

should also include any graphs or figures from the paper that would be helpful for us to look at as a group. Lastly, you should think of a few points to raise for discussion.

## Toddler Development Center reflection (10%)

We'll visit the Barnard Center for Toddler Development at some point in the semester to observe children participating in experiments or in structured play. Your task during the 30-minute visit is to pick a child and transcribe what they say. After the visit, you'll be asked to write a short, guided reflection about what you saw (see the assignment on CourseWorks for more information).

## Final project (60%)

The final project is a chance to apply what you've learned to a problem or area that you find particularly interesting. It should take the form of an experimental proposal to address some open question in some area of language development. Any of the methods that we discuss in class can be the basis of your proposed experiment. Depending on your question, you might, for example, propose a habituation-switch task with infants, a preferential looking task with older babies, a truth-value judgement task with toddlers, a corpus analysis of child directed speech, or a human simulation paradigm experiment with adults.

Your topic can be a follow-up to one of the topics we discussed in class or it can be centered on a completely new domain that we didn't discuss. The textbook might provide some ideas on this front (since we won't cover every topic in there, let alone every topic in the field). You might also have a favorite linguistic phenomenon for which you want to ask: when and how do children learn the knowledge governing *that*? (If you don't have a favorite phenomenon, any intro syntax textbook is a great starting point.)

The final project will be broken into sections (the due dates for each are listed in the course schedule below):

**Topic pitch (5%)**

A brief (~paragraph) outline of what sort of thing you're thinking about doing for the final project. You don't yet need a concrete idea; I just want to know the general domain / kind of thing you find interesting or think you want to read more about.

**Project proposal (10%)**

A ~500 word proposal outlining the question you'll be targeting, specifying the hypothesis or hypotheses you'll be testing, and the method you think will be most appropriate.

**Annotated bibliography (15%)**

A list of at least 5 peer-reviewed journal articles that you will use as background for the final paper. For each article, write a paragraph summarizing the paper's contribution and its relevance to your question. If it's an empirical paper, include information about what they found and the argument they made based on that data. If it's a review paper, outline their position on the issue. If it's a "theoretical" paper introducing a phenomenon, describe, in your own words, what knowledge the child needs to acquire to show adult-like mastery of that phenomenon.

**Final paper (30%)**

A 2,000-2,500 word paper that raises your question, summarizes the relevant background literature and identifies the gap your project plans to fill, then outlines the proposed experiment that will do so.

### Participation (15%)

Students are expected to attend and participate actively in class. This requires thoroughly reading and digesting the assigned readings for that day ahead of time, and being prepared to discuss your thoughts and questions about those readings in class. In particular, come prepared to identify what you liked about the papers and what you disliked or found unclear. Think about how you might connect each reading to that week's topic as well as to earlier material we discussed in class. Speak up when you are unconvinced by the logic of an argument or have a different argument that you would like to propose. That said, you shouldn't feel that you need to be absolutely confident in an idea to bring it up! All discussion is of course expected to be respectful of the different identities, backgrounds, and perspectives of your fellow students.

Participating in this sort of class is valuable for everyone. It greatly increases your understanding of the material, it benefits your classmates who may have had the same question or may benefit from hearing a different perspective from their own, and it benefits me in tailoring the course material to better serve your learning needs.

I'll use this grading scale for participation (out of 15 points):

15 Consistently engaged
10 Occasionally engaged
5 Often absent / not paying attention
0 Never attends
0 Rude or dismissive behavior

### Course Grading Scale

| Letter/Range | | | |
|---|---|---|---|
| A+ 99–100% | B+ 87–89% | C+ 77–79% | D 60–69% |
| A 93–98% | B 83–86% | C 73–76% | F Below 60% |
| A- 90–92% | B- 80–82% | C- 70–72% | |

Numerical grades will be rounded to the nearest integer before conversion to a letter grade (e.g., 89.5% ⇒ 90% ⇒ A-; 89.49% ⇒ 89% ⇒ B+). Final grades will not be curved.

## 4. Course Schedule

The schedule below is subject to change (any updates will be reflected on the class CourseWorks page, so check there for the most current version). Readings labeled as "L&K Chapter X" are from Lidz & Knowlton (2026) *A course in first language acquisition*. Other readings are journal articles and can be found on CourseWorks. Readings listed below are always due before class on the day that they're listed; Assignments are due that day at midnight anywhere on earth. FYI the Add deadline is Sept. 18th & the Drop deadline is Oct. 13th.

| **Date** | **Topic** | **Readings** (due before class) | **Assignments** |
|---|---|---|---|
| Tues, Sept 8 | What Is a Grammar, How Could You Learn One, And How the Heck Can We Study That? | | |

| Date | Topic | Readings (due before class) | Assignments |
|---|---|---|---|
| Tues, Sept 15 | What Is Learning? Data- vs. Knowledge-Driven Approaches | **L&K Chapters 2-3** [pp. 14-34]<br>**Gallistel (2010)** [pp. 1-10]<br>**Amichai & Yovel (2021)** [pp. 1-6] | |
| Tues, Sept 22 | How Does Input Matter? Inductive vs. Abductive Inference | **L&K Ch. 4** [pp. 36-42]<br>**Bergelson et al. (2023)** [pp. 1-9]<br>**Bergelson (2020)** [pp. 1-8]<br>**Senghas, Kita, & Özyürek (2004)** [pp. 2-5] | |
| Tues, Sept 29 | Another Point for Abduction: The Poverty of the Stimulus | **L&K Chapter 5** [pp. 44-54]<br>**Crain & Nakayama (1987)** [pp. 2-22]<br>**Legate & Yang (2002)** [pp. 1-11] | **Sign up to present (any paper with "←")** |
| Tues, Oct 6 | Word Learning: Association vs. Fast-Mapping | **L&K Chapter 6** [pp. 56-67]<br>**Smith & Yu (2008)** [pp. 1-9] ←<br>**Gleitman & Trueswell (2020)** [pp. 1-22] ← | |
| Tues, Oct 13 | Categorizing Sounds: Phonetics & Phonology | **L&K Chapter 7** [pp. 71-83]<br>**Werker (1995)** [pp. 1-17]<br>**Maye, Werker, & Gerken (2002)** [pp. 1-9] ←<br>**Yeung & Werker (2009)** [pp. 1-9] ← | **Topic pitch** |
| Tues, Oct 20 | Syntactic Categories I: The Determiner Productivity Debate | **L&K Chapter 8** [pp. 85-91]<br>**Valian, Solt, & Stewart (2009)** [pp. 1-31] ←<br>**Yang (2013)** [pp. 1-4] ←<br>**Shi & Melançon (2010)** [pp. 1-14] ←<br>**Babineau, Shi, & Christophe (2020)** [pp. 1-13] ← | |
| Tues, Oct 27 | Syntactic Categories II: Nouns, Adjectives, Verbs, & Corresponding Conceptual Categories | **L&K Chapters 9-10** [pp. 93-111]<br>**Waxman et al. (2009)** [pp. 1-22] ←<br>**He & Lidz (2017)** [pp. 2-21] ← | |
| Tues, Nov 3 | NO CLASS (election day) | | |
| Tues, Nov 10 | Learning (a bit about) Meanings: Syntactic Bootstrapping | **L&K Chapter 11** [pp. 112-126]<br>**Yuan & Fisher (2009)** [pp. 1-7] ←<br>**Lidz & Gleitman (2004)** [pp. 1-5] ← | **Project proposal** |
| Tues, Nov 17 | Learning (some) Syntax: Semantic & Prosodic Bootstrapping | **L&K Chapter 14-15** [pp. 159-174]<br>**Gout, Christophe, & Morgan (2004)** [pp. 1-17] ←<br>**de Carvalho et al. (2019)** [pp. 1-12] ← | |
| Tues, Nov 24 | NO CLASS (Happy Thanksgiving!) | | |

| Date | Topic | Readings (due before class) | Assignments |
|---|---|---|---|
| Tues, Dec 1 | Hierarchy: Quantifier Scope | **L&K Chapter 13** [pp. 145-158]<br>**Lidz & Musolino (2002)** [pp. 1-34] ←<br>**Viau, Lidz, & Musolino (2010)** [pp. 1-22] ← | **Annotated bibliography** |
| Tues, Dec 8 | Dependencies: Wh-movement & Islands | **L&K Chapters 17-18** [pp. 193-213]<br>**Perkins & Lidz (2021)** [pp. 1-5] ←<br>**Fetters & Lidz (2019)** [pp. 1-14] ← | |
| Tues, Dec. 22 | | | **Final paper** |

## 5. Course Policies

### Anonymous Feedback

I'm always interested in improving my teaching and my classes, so if there's any feedback you'd like to share during or after the course, you can do so anonymously with this form.

### Late Policy

Work submitted late without a prior extension will be marked down 10% if submitted within 24 hours of the deadline, and an additional 5% for each additional day late. If you know in advance that, for whatever reason, you'll need an extension on some aspect of the final project, email me!

### Barnard College Honor Code

Approved by the student body in 1912 and updated periodically, the Honor Code states:

*"We, the students of Barnard College, resolve to uphold the honor of the College by engaging with integrity in all of our academic pursuits. We affirm that academic integrity is the honorable creation and presentation of our own work. We acknowledge that it is our responsibility to seek clarification of proper forms of collaboration and use of academic resources in all assignments or exams. We consider academic integrity to include the proper use and care for all print, electronic, or other academic resources. We will respect the rights of others to engage in pursuit of learning in order to uphold our commitment to honor. We pledge to do all that is in our power to create a spirit of honesty and honor for its own sake."*

More details (including 2026 updates) can be found at: https://barnard.edu/student-conduct-pilot

### Use of Generative AI Tools

For the current academic year (2026–2027), Barnard College is providing an AI Usage Scale (see below) to allow faculty to indicate whether AI use is acceptable in a given course and, where applicable, the level and type of AI permitted for specific assignments. **The scale number 1 applies for everything in the course.** That is, you may consult AI while drafting your presentation slides / organizing your notes or when looking for papers potentially related to your topic of interest. You can even bounce ideas off of AI if you think that would be helpful. In all cases though, if you use AI, you should acknowledge that you used it and say exactly how (e.g., include your prompts in your submission). In any case, the final paper should be your own. Please feel free to consult with me if at any point you have questions about the kinds of secondary material and tools allowed for this course.

| | |
|---|---|
| **0 - No AI** | You must not use AI at any point during the assignment. You must demonstrate your own skills and knowledge. |
| **1 - AI Planning** | You may use AI for planning, idea development, and research. Your final submission should show how you have developed and refined these ideas. |
| **2 - AI Collaboration** | You may use AI to assist with specific tasks related to refining your original work. As part of your process, you should critically evaluate and modify any AI-generated content you use. |
| **3 - Full AI** | You may use AI extensively throughout your work either as you wish, or as specifically directed in your assignment. The goal here is strategic use of existing AI tools to augment your own work. |
| **4 - AI Exploration** | You should use AI creatively to solve the task, working to develop new tools and extending our understanding of what AI can do. |

## Class Climate

Seminars on language acquisition and cognitive science can raise questions that people can reasonably disagree about, and while lively discussion is welcome, please engage each other's ideas generously and respectfully; aggressive, dismissive, or intolerant speech toward classmates is not acceptable. If you ever have concerns about how you or a classmate has been treated in this course, please reach out to me.

## Center for Accessibility Resources & Disability Services (CARDS)

If you anticipate barriers to your academic experience due to a documented disability or emerging health challenge, please contact your instructor and/or the Center for Accessibility Resources & Disability Services (CARDS) as early as possible. If you have questions regarding registering a disability or receiving accommodations for the semester, contact CARDS at (212) 854-4634 or cards@barnard.edu. You can learn more about on-campus support at barnard.edu/disability-services. CARDS is located in 307 Diana Center.

## Wellness Statement

It is important for you as undergraduates to recognize and identify the different pressures, burdens, and stressors you may be facing, whether personal, emotional, physical, financial, mental, or academic. As a community, we urge you to prioritize yourself — your health, sanity, and wellness — throughout your career on campus. Sleep, exercise, and eating well are all part of a healthy regimen to cope with stress. Resources exist to support you, and we encourage you to make use of them. Should you have any questions about navigating these resources, please visit these sites:

- http://barnard.edu/primarycare
- https://barnard.edu/about-counseling
- https://barnard.edu/TheWellnessSpot/about

## Affordable Access to Course Texts & Materials Statement

All students deserve to be able to study and make use of course texts and materials regardless of cost. Barnard librarians have partnered with students, faculty, and staff to find ways to increase student access to textbooks. By the first day of advance registration for each term, faculty are expected to provide information about required texts for each course on CourseWorks (including ISBN or author, title, publisher, copyright date, and price), which can be viewed by students. A number of cost-free or low-cost methods for accessing some types of course texts are detailed in the Barnard Library Textbook Affordability Guide (library.barnard.edu/textbook-

affordability). Undergraduate students who identify as first-generation and/or low-income may check out items from the FLIP lending libraries in the Barnard Library (https://library.barnard.edu/find-borrow/FLIpartnership) and in Butler Library for an entire semester. Students may also consult with their professors, librarians, the Dean of Studies, and the Financial Aid Office about additional affordable alternatives for access to course texts.